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The lecture-hall quadratic assignment problem

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ABSTRACT

We study the post-timetabling *lecture-hall assignment problem*: given a fixed weekly timetable and realized student enrollments, assign each lecture to a compatible hall so as to minimize the walking burden of students moving between consecutive lectures, plus a penalty for feasible but non-ideal hall-lecture assignments. The problem combines a quadratic assignment objective, driven by shared enrollments and hall-to-hall distances, with interval-scheduling feasibility and hall-capacity constraints. We prove that it is NP-hard and, unless P = NP, inapproximable within any fixed rational factor, already in a single-day, two-block special case. We develop two exact mathematical-programming formulations: a bilinear mixed-integer quadratic program and a compact mixed-integer linear program with one distance variable per successor pair. To tighten the models we introduce biclique distance cuts that dominate the standard pairwise linking inequalities, complemented by Chvátal-Gomory capacity-dominance cuts; common side constraints such as *SameRoom* and *SameAttendees* are accommodated as well. In experiments on 35 daily instances derived from five large ITC 2019 competition benchmarks and a 2023 institutional dataset from Lancaster University, the biclique-strengthened compact linearized program solves every instance to proven optimality within an 1800-second limit. The direct quadratic model often finds the same incumbents but is weaker for proving optimality on the hardest instances, while the biclique cuts are the dominant modeling device behind the observed speed and robustness. On the ten Lancaster instances, optimized assignments reduce model-implied walking by 20%–45% relative to the deployed room plan.

1. Introduction and background

Universities and colleges often determine the course timetable well before they finalize the assignment of lectures to specific halls. Throughout this paper, we use the term *hall* (or *lecture-hall*) to refer generically to any space used for academic activity, whether it is a traditional lecture theater, a seminar room, or a laboratory. Similarly, we use the term *lecture* to refer to any scheduled academic activity, encompassing regular lectures, tutorials, or practical sessions. Once the timetable is fixed and student registration is completed, the institution knows both the enrollment in each lecture and the enrollment overlap between consecutive lectures. At that stage, the hall-assignment decision can have a substantial effect on student mobility across campus. Assigning consecutive lectures attended by many of the same students to distant halls may generate unnecessary walking, even when every lecture is assigned to a feasible hall. In practice, long walks between halls may also cause students to arrive late at their next lecture, thereby disrupting the beginning of a lecture. Such walks can be particularly challenging for students with mobility disabilities.

This observation motivates treating lecture-hall assignment separately from timetabling. In the lecture-hall assignment problem, the timetable, including the scheduled time interval of each lecture, is given. The remaining task is to assign lectures to halls in a way that respects hall compatibility in terms of capacity and required facilities, while also reducing the walking burden imposed on students as they move between consecutive lectures. Other, more local, considerations may also affect the desirability of assigning each lecture to a compatible hall; for example, large halls are feasible but less desirable for lectures with low enrollment.

Although hall assignment and time scheduling are intrinsically connected, and are indeed often modeled jointly in the timetabling literature, the two decisions are separated in practice at many institutions, for structural reasons. The timetable must be published well before the term starts so that students can plan and register and teaching staff can be recruited, whereas reliable enrollment figures, the very data that drive the walking-burden objective, become available only after registration closes. At this late stage, the institution can also accommodate mobility needs identified among enrolled students, for example by assigning their lectures to wheelchair-accessible halls. Moreover, reassigning halls shortly before or even during the term is operationally cheap, whereas changing lecture times at that stage disrupts students

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Table 1
Related literature, stream 1: the quadratic assignment problem and its solution machinery. "Type" indicates whether the work develops an exact method, a bound, or a heuristic.

Reference	Main device	Type	Relation to the present work
Koopmans and Beckmann (1957)	Problem definition	–	Origin of the flow–distance objective structure
Gilmore (1962) and Lawler (1963)	Gilmore–Lawler bound	Bound	Classical baseline relaxation
Kaufman and Broeckx (1978)	Compact linearization	Exact	Template for our compact distance-variable MILP
Frieze and Yadegar (1983)	Permutation-based linearization	Exact	Alternative linearization
Adams and Johnson (1994)	Disaggregated reformulation	Exact	The dense alternative our compact model avoids
Taillard (1991)	Robust tabu search	Heuristic	Candidate warm start (future work)
Hahn and Grant (1998)	Hungarian-based dual bound	Bound	Stronger-bound alternative
Zhao et al. (1998)	Semidefinite relaxation	Bound	Stronger-bound alternative (see Section 3.3)
Anstreicher et al. (2002)	Branch-and-bound on grids	Exact	Scale reference for pure QAP
Anstreicher (2003)	Convex quadratic reformulation	Exact	Possible tighter relaxation of the walking term
Burer and Vandenbussche (2006)	Lift-and-project branch-and-cut	Exact	Inspiration for the biclique strengthening
Fischer et al. (2019)	Layered MIP/SDP branch-and-cut	Exact	Inspiration for the layered linearization

and instructors. The hall-assignment stage is therefore naturally resolved late, when the timetable is already immutable. At institutions that assign halls jointly with the timetable, the same model can serve as a second-stage repair once actual enrollments are revealed.

Analogous post-scheduling location-assignment problems arise in other service systems in which the schedule is fixed first and the physical resource is assigned later. In airport operations, gates are assigned to flights of an already published schedule, classically with the objective of minimizing passenger walking distances between connecting flights (Haghani and Chen, 1998); in hospitals, operating rooms are allocated to a fixed surgical case schedule (Cardoen et al., 2010); and examination venues are commonly assigned only after the exam timetable and registration counts are known. The lecture-hall assignment problem studied here belongs to this family, with the distinctive feature that the coupling between consecutive activities is quadratic, driven by shared attendance and inter-hall distances. In this paper, we model this problem and present exact solution methods for it.

The proposed model is related to three main streams of literature: the quadratic assignment problem (QAP), lecture-hall assignment and interval scheduling, and university course timetabling. The closest conceptual link to the QAP is the presence of pairwise interaction costs: assigning two lectures to two halls creates a cost that depends jointly on the distance between the halls and on the number of students who attend both lectures. This is analogous to the flow–distance structure that defines the classical QAP introduced by Koopmans and Beckmann (1957) and surveyed extensively by Burkard et al. (1998) and Loiola et al. (2007). The QAP literature has developed a rich toolkit of bounds, linearizations, and exact and heuristic algorithms whose ideas we draw on in this paper; Table 1 summarizes the works most relevant here and the role each plays in the present paper, from the classical Gilmore–Lawler bound (Gilmore, 1962; Lawler, 1963) and the compact linearizations of Kaufman and Broeckx (1978) and Frieze and Yadegar (1983), through stronger disaggregated reformulations (Adams and Johnson, 1994) and dual, semidefinite, and convex-quadratic bounds (Hahn and Grant, 1998; Zhao et al., 1998; Anstreicher, 2003), to lift-and-project branch-and-cut (Burer and Vandenbussche, 2006), large-scale exact computation (Anstreicher et al., 2002), layered branch-and-cut for related facility-layout problems (Fischer et al., 2019), and robust tabu search (Taillard, 1991). The compact linearization and the valid inequalities that we develop in Section 3 are inspired by this branch-and-cut line of work, in particular by the lift-and-project ideas of Burer and Vandenbussche (2006) and the layered linearization of Fischer et al. (2019), but are adapted to the combined temporal and compatibility structure of the lecture-hall setting and exploit the sparsity of the successor relation that is absent from the dense classical QAP. At the same time, the present problem differs from the standard QAP because feasibility is driven by temporal overlap, hall capacities, and the subset of lecture pairs that are consecutive on the same day.

A second relevant stream concerns lecture-hall assignment and interval-based hall allocation. In this family of problems, activities with fixed or partially fixed time intervals must be assigned to halls while respecting incompatibilities caused by overlap in time. Carter and Tovey (1992) study the lecture-hall assignment problem from a complexity perspective and explicitly connect it to interval scheduling. The broader interval-scheduling literature, surveyed by Kolen et al. (2007), studies machine or resource assignment of activities with fixed time windows and provides a natural framework for the temporal-feasibility layer of our problem. In particular, the clique structure of interval graphs, which we exploit in Section 3 to tighten the overlap constraints, is a standard building block in that literature. The framework of Carter and Tovey (1992) and Kolen et al. (2007) is closely related to the feasibility component of the present model, where lectures cannot share a hall if their time intervals overlap. The present objective additionally includes interaction costs across consecutive lectures through student flows and hall-to-hall distances.

A third stream is the university timetabling literature, which studies the joint allocation of courses to times and halls under institutional constraints. The literature refers to this problem as the *University Course Timetabling Problem*. General surveys by Schaerf (1999) and Burke and Petrovic (2002) describe a broad range of formulations, including problems in which hall capacities, conflicts, and soft preferences play a central role, while the more recent review of Abdipoor et al. (2023) summarizes modern meta-heuristic and hybrid approaches for university course timetabling. The field has also been shaped by three international timetabling competitions (Lewis and Paechter, 2005; McCollum et al., 2010; Bonutti et al., 2012; Abdullah and Turabieh, 2012), whose formulations, benchmarks, and representative solution approaches are summarized in Table 2; the ITC 2019 competition (Müller et al., 2025) generalized the earlier models with rich distribution constraints and student sectioning, and produced the dataset used in our real-world campaign. Within this literature, lecture-hall assignment is often treated either as a subproblem of course timetabling or as a second-stage problem after times have been fixed. For example, Torres-Ovalle et al. (2014) propose a two-phase approach in which course scheduling is followed by lecture-hall assignment, and Rappos et al. (2022) develop a mixed-integer programming model that jointly considers student allocation, hall assignment, and time-period assignment. A recent extension by Davison et al. (2025) brings hybrid teaching into the university course timetabling problem, treating each lecture's delivery mode as a first-class element of a multiobjective binary program solved lexicographically on institutional data; like most models in this tradition, however, it keeps hall assignment embedded inside the larger multi-criterion formulation rather than isolating it as a separate problem with exploitable structure. This separation is natural in many university settings and is also compatible with the present formulation, which assumes that the scheduled time interval of each lecture is already known.

Table 2
Related literature, streams 2 and 3: lecture-hall assignment, interval-based hall allocation, and university course timetabling. "UCTT" denotes university course timetabling, and "Post-TT" denotes post-timetabling hall assignment under a fixed schedule. The last row positions the present paper.

Reference	Setting	Objective(s)	Method	Data
Carter and Tovey (1992)	Post-TT hall assignment	Feasibility, hall utilization	Complexity analysis	–
Lewis and Paechter (2005)	Joint UCTT	Feasibility, soft preferences	Graph-based heuristics	ITC 2002
McCollum et al. (2010)	Joint UCTT	Soft-constraint penalties	Competition overview	ITC 2007
Bonutti et al. (2012)	Joint curriculum-based UCTT	Soft-constraint penalties	Benchmark protocol	CB-CTT
Abdullah and Turabieh (2012)	Joint UCTT	Soft-constraint penalties	Memetic heuristic	CB-CTT
Torres-Ovalle et al. (2014)	Two-phase: times, then halls	Hard/soft constraints; hall utilization	Hierarchical two-phase ILP	U. La Sabana (2012)
Rappos et al. (2022)	Joint (students, halls, times)	Linear	MIP (exact)	Institutional
Müller et al. (2025)	Joint UCTT	Distribution-constraint penalties	Competition overview	ITC 2019
Davison et al. (2025)	Joint UCTT with hybrid teaching	Multiobjective, lexicographic	Binary program	Lancaster
Antunes-Batista et al. (2026)	Post-TT hall allocation	Six criteria, multiobjective	MOEA + local search; ILP baseline	Institutional (unnamed)
This paper	Post-TT hall assignment	Quadratic walking cost + linear penalties	Exact MIQP/MIP with valid inequalities	ITC 2019 + Lancaster

Taken together, these three lines of literature suggest that the lecture-hall assignment problem studied here combines structural ingredients that are usually treated separately. From lecture-hall assignment and interval scheduling it inherits the temporal packing constraints on halls; from timetabling it inherits the academic context, capacity restrictions, and interpretation of lectures as scheduled activities; and from the QAP it inherits a pairwise objective that couples assignment decisions across lectures. A particularly close recent connection is the hall-allocation study of Antunes-Batista et al. (2026), who isolate the hall-assignment subproblem of the university course timetabling problem under an already-fixed weekly schedule and formulate it as a six-criterion problem with two hard constraints (hall capacity and no scheduling overlap) and four soft objectives (student relocation between buildings, mismatch between requested and assigned hall types, wasted capacity, and hall changes between consecutive lectures of the same student group). They develop customized multiobjective evolutionary algorithms augmented with problem-specific local search and constructive heuristics implemented in the jMetal framework, and benchmark them against an ILP model solved by Gurobi. The computational study uses a full-semester dataset provided by an unnamed university, with approximately 10,000 students, more than 26,000 scheduled lecture sessions, and over 120 lecture halls of more than 20 distinct types, and emphasizes scalability and Pareto-front quality as the central performance measures.

The present framework can likewise accommodate mandatory hall features through the compatibility sets and additional hall-quality criteria through linear assignment penalties, so on the modeling surface it overlaps with four of the six criteria of Antunes-Batista et al. (2026). Its distinguishing features are structural and methodological. Structurally, student movement enters as a QAP-style quadratic term induced by shared enrollments and inter-hall walking distances rather than as a count of building changes or group-level hall changes; capacity and overlap are retained as hard feasibility constraints; and the remaining hall-quality criteria are combined into a single scalar objective through linear assignment penalties, which is possible because the weighting between them is under the analyst's control. Methodologically, this combined structure motivates exact mathematical-programming approaches with problem-specific valid inequalities rather than the multiobjective evolutionary framework favored in Antunes-Batista et al. (2026). Their approach targets very large operational instances where a Pareto front among competing criteria is the primary deliverable, whereas ours targets a QAP-flavored subproblem in which pairwise structure and interval-graph cliques support exact optimization.

The relation between the two solution frameworks can be made precise. The exact component of Antunes-Batista et al. (2026) is a single weighted-sum integer linear program in which the two hard requirements, capacity and non-overlap, are themselves penalized objectives, supplemented by a dummy room that guarantees feasibility. The present model instead combines a quadratic walking term with general inter-hall distances and shared-enrollment weights, a compact distance-variable linearization, biclique and capacity-dominance inequalities, and clique-based hard non-overlap constraints. Some of this machinery transfers to their setting. Their student-relocation and room-change criteria are special cases of our walking term in which the distance matrix is the 0/1 indicator of a building or room change and every qualifying class pair carries unit weight, so the compact linearization and the biclique cuts apply verbatim and could, for example, provide provably optimal anchor points for their Pareto analysis. Their pairwise criteria range only over consecutive sessions of the same course unit within the same shift, so they capture a cohort staying with its own course, whereas our successor relation is derived from individual registrations and couples lectures of different courses whenever students move between them. These cross-course flows constitute the bulk of the walking burden on our benchmarks: successor pairs whose lectures belong to different courses carry between 75% and 100% of the total shared-enrollment weight on the 35 benchmark days used in Section 4, 91% on average. The two objectives therefore measure materially different quantities even before weights and distances are considered. The studies also use different feasibility conventions: capacity and overlap violations are penalized there but are hard constraints here. Because the article provides no public dataset or repository link, the published materials do not support a like-for-like computational comparison.

Table 2 summarizes the hall-assignment, interval-scheduling, and timetabling streams, and positions the present work within them.

Research gap and contribution. While the QAP literature provides powerful exact algorithms (Table 1), it assumes unrestricted assignments without temporal overlap or capacity constraints. Conversely, the timetabling and interval-scheduling literatures address these operational realities but typically rely on linear objectives or multiobjective heuristics rather than QAP-style quadratic pairwise interactions (Table 2).

We bridge this gap by isolating the post-timetabling lecture-hall assignment problem, a setting common in institutions where course enrollments are only revealed after the schedule is set. In our framework, temporal overlap and spatial capacity act as hard constraints (with optional linear penalties for soft preferences), while the primary objective is a quadratic mapping of walking effort driven by shared student enrollments. By formally separating this hall-assignment stage from the surrounding timetabling process, we exploit its distinct structure. We analyze its computational complexity and construct a compact two-index distance-variable formulation in the spirit of classical compact QAP linearizations. To strengthen this model, we introduce easy-to-separate biclique inequalities that aggregate the standard pairwise links over compatible hall subsets, drawing on classical lift-and-project ideas while exploiting the sparsity of lecture-hall compatibilities. Through comprehensive numerical studies, we demonstrate that these problem-specific mathematical-programming enhancements substantially improve computational performance over the direct nonlinear formulation, solving large, realistic instances to optimality within practical time limits.

2. Problem statement and complexity

We now formalize the lecture-hall assignment problem and state its complexity properties. The problem is defined through the members of the following sets:

Halls (H) Each hall $h \in H$ is defined by its capacity u_h, its location, and its set of available features F_h (such as "laboratory bench", "computer workstation", "projector", or "wheelchair-accessible entrance").

Lectures (L) Each lecture $l \in L$ is defined by its scheduled time interval, its number of registered students s_l, and its set of mandatory hall features F_l^{req} (for example, a chemistry practical must be assigned to a lab hall, and a programming session must be assigned to a hall with computer workstations).

Immediate-successor relation (A) A relation $A \subseteq L \times L$, where $(l_1, l_2) \in A$ if there exists at least one student enrolled in both lectures such that, in that student's schedule, l_2 immediately follows l_1 on the same day.

A hall h is *compatible* with a lecture l if it has sufficient capacity and provides every mandatory feature, that is,

$$H(l) = \{h \in H : u_h \geq s_l,\ F_l^{\text{req}} \subseteq F_h\}.$$

The compatibility set $H(l)$ is computed once from the input data and is treated as primitive in the rest of the paper. Mandatory features are handled as hard restrictions through $H(l)$ because, for example, a chemistry practical cannot take place in a hall without lab benches regardless of any student-flow gain.

The walking distance between each pair of halls $h_1, h_2 \in H$ is given by the distance matrix d_{h_1,h_2}. In addition, each pair $(l_1, l_2) \in A$ has a positive shared-enrollment weight $c_{l_1,l_2} > 0$ measuring the students for whom l_2 is the next lecture after l_1. This quantity may be a simple head count or a score that gives higher weight to students with disabilities, if the institution has this information at its disposal. Finally, for each compatible lecture–hall pair (l, h) with $h \in H(l)$, we are given a nonnegative *assignment penalty* $p_{lh} \geq 0$ that captures local hall-quality considerations beyond the hard compatibility filter. In practice, p_{lh} is constructed by aggregating several soft per-assignment effects: wasted seating capacity when a small lecture is placed in a much larger hall, absence of desirable but non-mandatory features (for example, an oversized whiteboard desired by the instructor or a preferred building wing), and institutional preferences inherited from a higher-level timetabling stage. Any such weighting is encoded by adjusting the input p_{lh} before solving and does not change the formulations developed below; the relative importance of each component is therefore under the analyst's control. Together, the three ingredients d_{h_1,h_2}, c_{l_1,l_2}, and p_{lh} capture the cross-lecture coupling driven by student movement and the per-assignment local quality of each hall choice.

We make the following standing assumption on the registration data.

Single-lecture-at-a-time assumption No student is registered for two lectures whose scheduled intervals overlap in time.

Under this assumption, every student contributes to the shared enrollment c_{l_1,l_2} of a successor pair $(l_1, l_2) \in A$ at most once, and the walking-distance objective defined in the next section is well-posed on every feasible assignment. Any input that violates the assumption must be cleaned beforehand by dropping the conflicting registration from one of the two overlapping lectures; Section 4.1 describes one such cleaning step in our Lancaster benchmark.

The goal is to assign each lecture to a compatible hall such that no two lectures assigned to the same hall overlap in time. The objective is to minimize the sum of two terms: (i) the total walking distance traveled by students between consecutive lectures, and (ii) the total assignment penalty incurred by the chosen lecture-hall pairs. The relative weight between the two terms is implicit in the units of c_{l_1,l_2} and p_{lh}, and is therefore controlled by the analyst when these inputs are constructed.

A natural decision version of the lecture-hall quadratic assignment problem asks whether, for a given bound B, there exists a feasible hall assignment whose total cost is at most B. This decision problem belongs to NP, since a proposed assignment can be verified in polynomial time by checking compatibility, non-overlap, and the resulting objective value. The next proposition establishes its exact QAP-level hardness.

Proposition 1. *The decision version of the lecture-hall assignment problem is NP-complete, and the corresponding optimization problem is NP-hard. These statements remain true even for a restricted single-day case with two consecutive time blocks, identical hall capacities, and full lecture-to-hall compatibility.*

Proof. We use a reduction from the classical QAP restricted to zero-diagonal distance matrices, which remains NP-hard; see Burkard et al. (1998) and Loiola et al. (2007). Consider such a QAP instance with facilities $\mathcal{N} = \{1, \dots, n\}$, locations $\mathcal{R} = \{1, \dots, n\}$, nonnegative integer flow matrix (f_{ij}), and nonnegative integer distance matrix (d_{ab}) satisfying $d_{aa} = 0$ for every $a \in \mathcal{R}$. First normalize the distances by setting

$$\hat{d}_{aa} = 0 \quad \forall a \in \mathcal{R}, \qquad \hat{d}_{ab} = d_{ab} + 1 \quad \forall a \neq b.$$

For every permutation π of the locations,

$$\sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}\hat{d}_{\pi(i),\pi(j)} = \sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}d_{\pi(i),\pi(j)} + \sum_{i\neq j} f_{ij},$$

because $\pi(i) \neq \pi(j)$ whenever $i \neq j$ and the original diagonal entries are zero. Thus the normalization adds the same constant to every feasible QAP solution. It preserves exact comparisons, and a QAP decision threshold K is transformed into the threshold $K + \sum_{i\neq j} f_{ij}$.

Now construct a lecture-hall instance as follows. Create n halls $h_1, \dots, h_n$, all with identical capacity $u_{h_a} = U$ for some U larger than every lecture size introduced below, and declare the hall feature set $F_{h_a} = \emptyset$ for every a. For each lecture l introduced in the remainder of the construction, set the required-feature set $F_l^{\text{req}} = \emptyset$ and choose a registered-students count $s_l \leq U$, so that by the definition of compatibility in Section 2 every hall is compatible with every lecture. Set the assignment penalty $p_{lh} = 0$ for every lecture–hall pair so that the assignment-penalty term in the objective contributes nothing. Create two sets of lectures,

$$W = \{w_1, \dots, w_n\} \qquad \text{and} \qquad V = \{v_1, \dots, v_n\},$$

where all lectures in W occupy one common time block and all lectures in V occupy the immediately following common time block. Every lecture in W therefore overlaps every other lecture in W, and similarly for V. Since there are exactly n halls, every feasible lecture-hall assignment induces two permutations of the halls: one permutation π for the lectures in W and one permutation σ for the lectures in V. Set

$$d_{h_a,h_b} = \hat{d}_{ab} \qquad \forall a, b \in \{1, \dots, n\}.$$

Let M be a positive penalty parameter, to be specified below, and define

$$c_{w_i,v_j} = \begin{cases} f_{ij} + M, & \text{if } i = j, \\ f_{ij}, & \text{if } i \neq j. \end{cases}$$

Include (w_i, v_j) in A exactly when $c_{w_i,v_j} > 0$. Set $s_{w_i} = \sum_j c_{w_i,v_j}$ and $s_{v_j} = \sum_i c_{w_i,v_j}$, and choose the common hall capacity U at least as large as the maximum of these lecture sizes. The shared-enrollment counts are mutually consistent: they can be realized by taking, for each

pair with $c_{w_i,v_j} > 0$, a disjoint group of c_{w_i,v_j} students who attend w_i and then v_j. Every such student attends exactly one lecture in each block, so the single-lecture-at-a-time assumption holds. Under a feasible lecture-hall assignment corresponding to permutations (π, σ), the objective value is

$$\sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}\hat{d}_{\pi(i),\sigma(j)} + M\sum_{i=1}^{n}\hat{d}_{\pi(i),\sigma(i)}.$$

If $\pi \neq \sigma$, then there exists some i such that $\pi(i) \neq \sigma(i)$, and therefore $\hat{d}_{\pi(i),\sigma(i)} \geq 1$. Hence every such assignment has value at least M. On the other hand, if $\pi = \sigma$, then the penalty term vanishes and the lecture-hall objective becomes

$$\sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}\hat{d}_{\pi(i),\pi(j)},$$

which is exactly the normalized QAP objective.

For the exact reduction, start from the QAP decision problem with threshold K and set

$$M = \hat{K} + 1, \qquad \hat{K} := K + \sum_{i\neq j} f_{ij}.$$

Then any lecture-hall solution of value at most $\hat{K}$ must satisfy $\pi = \sigma$, and once $\pi = \sigma$ its value is exactly the normalized QAP value of the same permutation. Therefore the original QAP instance is a yes-instance if and only if the constructed lecture-hall instance admits a feasible assignment of cost at most $\hat{K}$. The quantity $\hat{K}$ is a sum of at most n^2+1 nonnegative integers taken directly from the QAP input, so its binary bit-length is polynomial in the bit-length of that input; consequently $M = \hat{K}+1$ and the resulting walking weights $c_{w_i,v_j} \in \{f_{ij}, f_{ij}+M\}$ all fit in polynomially many bits, and the full lecture-hall instance is produced in polynomial time from the QAP input. Hence the construction is a Karp (polynomial-time many-one) reduction, and the decision version of the lecture-hall assignment problem is NP-hard. Since it belongs to NP, it is NP-complete. Consequently, the optimization problem is NP-hard. □

The exact reduction above may use up to n^2 successor pairs and does not by itself transfer multiplicative approximation ratios, because its distance normalization adds a constant to every objective value. The next proposition establishes approximation hardness directly while also showing that sparsity does not help. As a directed graph, the successor relation is always acyclic because every arc points forward in time; sparsity notions such as degree and treewidth therefore refer to its underlying undirected graph. The result holds when the underlying undirected successor graph is a single cycle, so that every lecture is incident to at most two positive-weight successor pairs and the successor graph has treewidth two.

Proposition 2. *Unless* P = NP*, for every fixed rational $\rho \geq 1$ there is no polynomial-time ρ-approximation algorithm for the lecture-hall assignment problem, even on instances with two consecutive time blocks, identical hall capacities, full lecture-hall compatibility, zero assignment penalties, and a successor relation whose underlying undirected graph is a single cycle, in which every lecture is incident to exactly two positive-weight successor pairs.*

Proof. We reduce from the Hamiltonian-cycle problem on an undirected graph $\Gamma = (N, E)$ with $|N| = n \geq 3$, exploiting the fact that the quadratic assignment objective with a cycle-structured flow graph is the traveling-salesman objective (Koopmans and Beckmann, 1957; Sahni and Gonzalez, 1976). Create n halls with identical capacities and distances $d_{h_a,h_b} = 1$ if $\{a, b\} \in E$, $d_{h_a,h_b} = B$ if $a \neq b$ and $\{a, b\} \notin E$, and $d_{h_a,h_a} = 0$, where $B = \lceil \rho n \rceil + 1$. Create lectures $W = \{w_1, \ldots, w_n\}$ occupying one common time block and $V = \{v_1, \ldots, v_n\}$ occupying the immediately following block, with empty feature requirements and zero assignment penalties, so that every hall is compatible with every lecture. The registrations are chosen so that the only positive shared enrollments are $c_{w_i,v_i} = M$ and $c_{w_i,v_{i+1}} = 1$ for all i (indices modulo n), with $M = \lceil \rho(n - 1 + B) \rceil + 1$; this is realized by M students attending w_i and then v_i, and one student attending w_i and then v_{i+1}, so all enrollments equal $M + 1$ and the single-lecture-at-a-time assumption holds. As an undirected graph, the positive-weight pairs, all directed from the first block to the second, form the single cycle $v_1, w_1, v_2, w_2, \ldots, v_n, w_n, v_1$.

A feasible assignment consists of two bijections π and σ from W and V to the halls. If $\sigma(v_i) \neq \pi(w_i)$ for some i, the pair (w_i, v_i) alone contributes at least M, because every off-diagonal distance is at least 1. If instead $\sigma(v_i) = \pi(w_i)$ for all i, the gadget pairs contribute nothing and the total cost is $\sum_i d_{\pi(w_i),\pi(w_{i+1})}$, the length of the closed tour visiting the halls in the order $\pi(w_1), \ldots, \pi(w_n)$. Every closed tour either uses only E-edges, in which case it traces a Hamiltonian cycle of Γ and costs exactly n, or contains a non-edge and costs at least $n - 1 + B$. Hence, if Γ is Hamiltonian the optimal value is n, while otherwise every feasible solution costs at least $\min\{M, n - 1 + B\}$, and both quantities exceed ρn, since $n - 1 + B > \rho n$ and $M > \rho(n - 1 + B) > \rho n$. A ρ-approximation algorithm applied to a Hamiltonian instance must therefore return a solution of value at most ρn, which forces $\sigma = \pi$ and a tour without non-edges, exhibiting a Hamiltonian cycle of Γ; applied to a non-Hamiltonian instance it returns a value exceeding ρn. The algorithm thus decides Hamiltonicity. All constructed numbers are integers with size polynomial in n for fixed rational ρ, and taking $\rho = 1$ yields NP-hardness of the exact problem. □

Structure beyond worst-case hardness. Propositions 1 and 2 together delimit where the difficulty of the problem does and does not reside. First, restricting the compatibility structure alone cannot yield tractable subclasses: both reductions use identical hall capacities and full lecture-hall compatibility, so the hardness persists even for the simplest conceivable compatibility families, including the nested (laminar) families that arise when compatibility is driven by capacity only. Second, by Proposition 2, sparsity of the successor graph does not help either: bounded degree and even bounded treewidth of its underlying undirected graph leave the problem inapproximable. This bound is tight in the two-block setting, since if every lecture is incident to at most one positive-weight pair, so that the successor graph is a matching, assigning each successor the hall of its predecessor is feasible under full compatibility and attains walking cost zero. Third, the temporal structure is the real driver of complexity, and the boundary is sharp: with a single time block per day the successor relation is empty, the walking term vanishes, and the problem reduces to a linear assignment problem solvable in polynomial time, whereas two consecutive blocks already carry the full hardness of the QAP. Between these extremes, a positive counterpart emerges once the right graph is considered. For each lecture l, form the closed interval from its start time through the latest of its end time and the start times of all successors l' with $(l, l') \in A$, and let the *interaction graph* join two lectures whenever these extended intervals intersect. This graph contains every overlap edge and every successor pair. Being an interval graph, its treewidth equals its clique number minus one (Golumbic, 2004), that is, $c' - 1$, where c' is the peak number of lectures that are simultaneously active or awaiting one of their successors. The primal graph of the resulting constraint-optimization problem is a subgraph of this interaction graph: compatibility and assignment penalties are unary, non-overlap is enforced by pairwise inequalities along overlap edges, and walking costs live on successor edges. Standard dynamic programming over the interval-graph tree decomposition therefore solves the problem exactly in $|H|^{O(c')}$ time, with states assigning halls to the lectures in the current bag and enforcing distinct halls only among lectures that are simultaneously active. The problem is polynomial whenever this active-or-awaiting concurrency parameter is bounded, which is consistent with both reductions above, in which all $2n$ lectures are packed into two blocks and c' grows linearly with the instance size. In summary, it is not the treewidth of the successor graph that governs tractability (Proposition 2) but that of the combined interaction graph, and a systematic study of tractable subclasses and fixed-parameter algorithms for such parameters is left for future research (Section 5).

3. Models and algorithms

This section presents two complementary exact mathematical-programming views of the lecture-hall assignment problem. We begin with a natural mixed-integer quadratic programming formulation, then derive a compact mixed-integer linear programming reformulation. Both formulations optimize the same combined objective from Section 2: the quadratic walking term weighted by shared enrollments and hall-to-hall distances, plus the linear assignment-penalty term $\sum_l p_{l,a_l}$. After these two base formulations, we describe valid capacity-dominance inequalities and biclique cuts that can be layered on top of them.

3.1. Mixed-integer quadratic programming formulation

The base problem decomposes by day because temporal-overlap constraints and immediate-successor pairs never cross day boundaries. The common fixed-timetable side constraints described in Section 3.4 also preserve this decomposition when they involve lectures within one day; in particular, SameAttendees does not couple different days once the timetable is fixed. A hard or soft SameRoom requirement linking lectures on different days is the important exception: it couples their hall choices, so the affected days must be solved jointly. We therefore formulate one daily subproblem and suppress the day index: L denotes the lectures of that day, A the induced successor pairs, $G = (L, E)$ the interval overlap graph, and $\mathcal{C}$ the family of maximal cliques of G.

A natural mixed-integer quadratic programming (MIQP) formulation is then as follows. For each lecture $l \in L$ and each compatible hall $h \in H(l)$, let

$$x_{lh} = \begin{cases} 1 & \text{if lecture } l \text{ is assigned to hall } h, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

Recall that s_l is the number of students attending lecture l, u_h is the capacity of hall h, and $H(l) \subseteq H$ is the compatibility set defined in Section 2 that respects both the capacity bound $u_h \geq s_l$ and the mandatory-feature inclusion $F_l^{\text{req}} \subseteq F_h$. The problem can then be written as

$$\min \quad \sum_{(l_1,l_2)\in A} c_{l_1,l_2} \sum_{h_1\in H(l_1)} \sum_{h_2\in H(l_2)} d_{h_1,h_2}\, x_{l_1h_1} x_{l_2h_2} + \sum_{l\in L} \sum_{h\in H(l)} p_{lh}\, x_{lh} \tag{2}$$

$$\text{s.t.} \quad \sum_{h\in H(l)} x_{lh} = 1, \qquad \forall l \in L, \tag{3}$$

$$\sum_{l\in C:\, h\in H(l)} x_{lh} \leq 1, \qquad \forall C \in \mathcal{C},\ \forall h \in H, \tag{4}$$

$$x_{lh} \in \{0,1\}, \qquad \forall l \in L,\ \forall h \in H(l). \tag{5}$$

The objective has two parts. The first is quadratic: the walking cost of assigning a lecture to a hall depends on where the immediately succeeding lecture is assigned, and each contribution is scaled by the shared-enrollment weight c_{l_1,l_2}. The second is linear: each compatible lecture–hall pair (l, h) contributes its local assignment penalty p_{lh} whenever it is selected. The first set of constraints ensures that each lecture is assigned to exactly one compatible hall, and the second enforces non-overlap by requiring that, for every hall and every maximal clique of mutually overlapping lectures, at most one lecture in that clique may use the hall.

The clique family $\mathcal{C}$ can be enumerated efficiently. The overlap graph G of a single day is an interval graph, so every maximal clique consists of the lectures whose intervals contain some common time point, which can be taken to be the start time of one of its lectures. Consequently, there are at most $|L|$ maximal cliques, and a sweep over the sorted start and end times enumerates all of them in $O(|L| \log |L|)$ time, where the logarithmic factor is due solely to sorting (Golumbic, 2004). On the discrete time grid of our input, the implementation collects the active-lecture set at every occupied slot, removes duplicates, and discards every set properly contained in another; the remaining sets are exactly the maximal cliques. Its cost is proportional to the total number of lecture-slot incidences plus a maximality filter over at most one candidate set per occupied slot. The same family $\mathcal{C}$ is reused by both formulations and by the capacity-dominance cuts of Section 3.3.

3.2. Mixed-integer programming reformulation

A compact linearization can be obtained by introducing one nonnegative continuous variable $z_{l_1l_2}$ for each successor pair $(l_1, l_2) \in A$, representing the realized hall-to-hall distance induced by that pair. The linear assignment-penalty term from (2) is retained unchanged. The problem can then be reformulated as the mixed-integer linear program

$$\min \quad \sum_{(l_1,l_2)\in A} c_{l_1,l_2}\, z_{l_1l_2} + \sum_{l\in L} \sum_{h\in H(l)} p_{lh}\, x_{lh} \tag{6}$$

$$\text{s.t.} \quad \text{Constraints (3) and (4)}$$

$$z_{l_1l_2} \geq d_{h_1,h_2}\left(x_{l_1h_1} + x_{l_2h_2} - 1\right), \qquad \forall (l_1,l_2) \in A, h_1 \in H(l_1), h_2 \in H(l_2), \tag{7}$$

$$x_{lh} \in \{0,1\}, \qquad \forall l \in L,\ \forall h \in H(l), \tag{8}$$

$$z_{l_1l_2} \geq 0, \qquad \forall (l_1,l_2) \in A. \tag{9}$$

This reformulation is exact. The assignment-penalty term is unchanged from the MIQP model. For the quadratic walking term, each lecture is assigned to exactly one hall, so every pair (l_1, l_2) has exactly one compatible hall pair (h_1, h_2) for which $x_{l_1h_1} = x_{l_2h_2} = 1$; for that pair, constraint (7) yields

$$z_{l_1l_2} \geq d_{h_1,h_2}.$$

For every other compatible hall pair $(h_1', h_2') \neq (h_1, h_2)$, at least one of $x_{l_1h_1'}$ and $x_{l_2h_2'}$ equals zero, so the bracket $x_{l_1h_1'} + x_{l_2h_2'} - 1$ is nonpositive; combined with $d_{h_1',h_2'} \geq 0$, the right-hand side of (7) is then at most zero and the corresponding constraint is already implied by the nonnegativity bound $z_{l_1l_2} \geq 0$. Therefore, because the objective minimizes $\sum c_{l_1,l_2} z_{l_1l_2}$ with $c_{l_1,l_2} > 0$ and $z_{l_1l_2} \geq 0$, the optimal value of $z_{l_1l_2}$ is exactly the realized distance d_{h_1,h_2} associated with the halls assigned to l_1 and l_2. Multiplying by c_{l_1,l_2} in the objective recovers the original walking-cost contribution.

Compared with the standard disaggregated QAP linearization, which introduces one variable for each tuple (l_1, l_2, h_1, h_2), formulation (6)–(9) requires substantially fewer additional variables. Placing the shared-enrollment weight c_{l_1,l_2} in the objective rather than in the linking constraints also yields a better-scaled coefficient matrix while preserving the same optimal assignment. Its linking relaxation need not dominate the standard disaggregated relaxation. The formulation is a sparse, successor-pair-indexed relative of the compact QAP linearization of Kaufman and Broeckx (1978). Strengthening inequalities for this compact reformulation are described in Section 3.3.

3.3. Modeling improvements

Several enhancements can be layered on top of the two base formulations without changing the underlying problem. The improvements presented below apply to both the MIQP and MIP formulations; only the encoding differs, adapting to each modeling view's natural primitives: linear inequalities in the MIP and bilinear inequalities in the MIQP.

Capacity-dominance cuts. A first general improvement combines the hall-capacity structure with the clique-overlap restriction. For each maximal clique $C \in \mathcal{C}$ and each capacity threshold $\kappa \geq 0$, define the

sets

$$L_\kappa(C) = \{\, l \in C : s_l > \kappa \,\}, \qquad H_\kappa = \{\, h \in H : u_h > \kappa \,\}.$$

Every lecture $l \in L_\kappa(C)$ has $H(l) \subseteq H_\kappa$ by the capacity definition, so all of its assignment mass is placed on halls in H_κ. Summing the clique-overlap inequalities (4) over $h \in H_\kappa$ and isolating the contribution of the large lectures yields the *capacity-dominance cut*

$$\sum_{l \in C \setminus L_\kappa(C)} \sum_{h \in H(l) \cap H_\kappa} x_{lh} \;\le\; |H_\kappa| - |L_\kappa(C)|, \qquad \forall C \in \mathcal{C},\ \forall \kappa \ge 0. \tag{10}$$

Because $|L_\kappa(C)|$ and $|H_\kappa|$ depend only on the input data, and in particular are independent of the hall-assignment decisions, the cut is an explicit linear inequality in the assignment variables with a precomputed integer right-hand side. A negative right-hand side certifies that the clique cannot be assigned feasibly; otherwise, the inequality states that once the $|L_\kappa(C)|$ largest lectures of the clique have claimed their share of the $|H_\kappa|$ halls whose capacity exceeds κ, the remaining lectures in C may collectively occupy at most $|H_\kappa| - |L_\kappa(C)|$ of the large halls. Only threshold values κ that coincide with distinct hall capacities need to be instantiated, so the family contains at most $|\mathcal{C}| \cdot |\{u_h : h \in H\}|$ inequalities. The cut is a Chvátal–Gomory consequence of (3)–(4) and is therefore valid for every integer-feasible solution of the MIQP and MIP formulations alike. In both formulations it can be added directly as a linear inequality in the assignment variables.

Biclique cuts. The next family of improvements strengthens the distance linking between successor pairs. We present it first for the compact MIP reformulation and then extend it to the MIQP formulation. Fix a pair $(l_1, l_2) \in A$ and a pair of halls $h_1 \in H(l_1)$ and $h_2 \in H(l_2)$. Let $S_1 \subseteq H(l_1)$ and $S_2 \subseteq H(l_2)$ be such that $d_{h'_1,h'_2} \ge d_{h_1,h_2}$ for every $h'_1 \in S_1$ and $h'_2 \in S_2$. Then the following inequality is valid:

$$z_{l_1 l_2} \ge d_{h_1,h_2} \times \left(\sum_{h'_1 \in S_1} x_{l_1 h'_1} + \sum_{h'_2 \in S_2} x_{l_2 h'_2} - 1 \right), \qquad \forall (l_1,l_2) \in A, h_1 \in H(l_1), h_2 \in H(l_2). \tag{11}$$

If the assigned hall of l_1 lies in S_1 and that of l_2 lies in S_2, then both summations equal 1 and (11) forces $z_{l_1 l_2} \ge d_{h_1,h_2}$; this bound is valid because every pair in $S_1 \times S_2$ has distance at least d_{h_1,h_2}. Otherwise the right-hand side is nonpositive and the cut is implied by the nonnegativity of $z_{l_1 l_2}$. The basic constraints (7) are the special case $S_1 = \{h_1\}$ and $S_2 = \{h_2\}$; taking larger bicliques can yield stronger cuts that separate fractional solutions satisfying all pairwise constraints.

In practice we generate a family of patterns (S_1, S_2, δ) by the anchor-based procedure in Algorithm 1. For each anchor pair $(h_1, h_2) \in H(l_1) \times H(l_2)$ the algorithm sets the common threshold $\delta = d_{h_1,h_2}$, expands S_2 to every hall in $H(l_2)$ at distance at least δ from the anchor h_1, and then expands S_1 to every hall in $H(l_1)$ at distance at least δ from every element of S_2. The anchor itself always satisfies $h_1 \in S_1$ and $h_2 \in S_2$, so (11) evaluated at the returned triple is at least as strong as the base pair-distance constraint obtained from the same anchor. Distinct anchors often produce the same triple, which is why the generated patterns are deduplicated before any cut is posted. For every returned triple (S_1, S_2, δ) we add one inequality of the form (11) with d_{h_1,h_2} replaced by δ, so the total number of biclique cuts posted for a successor pair is bounded by $|H(l_1)| \cdot |H(l_2)|$ and in practice is substantially smaller because of the deduplication. In the MIQP formulation, where the auxiliary distance variables $z_{l_1 l_2}$ are absent, the same biclique pattern (S_1, S_2, δ) gives the bilinear valid inequality

$$\sum_{h'_1 \in S_1} \sum_{h'_2 \in S_2} d_{h'_1,h'_2}\, x_{l_1 h'_1} x_{l_2 h'_2} \;\ge\; \delta \left(\sum_{h'_1 \in S_1} x_{l_1 h'_1} + \sum_{h'_2 \in S_2} x_{l_2 h'_2} - 1 \right), \tag{12}$$

which is posted directly as a quadratic constraint: when the assigned halls of l_1 and l_2 lie in S_1 and S_2, respectively, the left-hand side equals the realized bilinear distance contribution of the pair and is at least δ by construction of the pattern; otherwise the right-hand side is nonpositive and the cut is implied by nonnegativity of the bilinear terms.

Algorithm 1: Biclique distance patterns for a successor pair $(l_1, l_2) \in A$.
Input: compatibility sets $H(l_1)$, $H(l_2)$, hall-to-hall distance matrix d.
Output: deduplicated set $\mathcal{P}(l_1, l_2)$ of patterns (S_1, S_2, δ).

```
P(l1, l2) ← ∅
for each (h1, h2) ∈ H(l1) × H(l2) do
    δ ← d_{h1,h2}
    S2 ← { h ∈ H(l2) : d_{h1,h} ≥ δ }
    S1 ← { h ∈ H(l1) : d_{h,h''} ≥ δ for every h'' ∈ S2 }
    P(l1, l2) ← P(l1, l2) ∪ {(S1, S2, δ)}
end for
return P(l1, l2)
```

The general family (11) can be separated exactly in polynomial time. Fix a fractional point x^*, a successor pair $(l_1, l_2) \in A$, and a threshold δ drawn from the set of distinct entries of the distance matrix. An inequality of the family with coefficient δ is most violated when $\sum_{h'_1 \in S_1} x^*_{l_1 h'_1} + \sum_{h'_2 \in S_2} x^*_{l_2 h'_2}$ is maximized subject to $S_1 \times S_2$ forming a complete bipartite subgraph (a biclique) of the bipartite graph on $H(l_1) \cup H(l_2)$ whose edges are the hall pairs at distance at least δ. These biclique vertex sets are the nondegenerate independent sets of the bipartite complement graph. Independent sets supported on only one side cannot yield a violated cut: by the assignment equations their weight is at most one, and with $z_{l_1 l_2} \ge 0$ they cannot violate an inequality of the form (11). A maximum-weight independent set of a bipartite graph can be computed by a single minimum-cut computation via Kőnig-type duality; after the resulting vertex set is checked for positive violation, it gives exact separation for the fixed threshold. It suffices to consider distance-matrix values: if a cut with some intermediate δ is violated, then its total selected assignment weight exceeds one, and replacing δ by the minimum distance over $S_1 \times S_2$, which is itself a distance-matrix entry, only strengthens the right-hand side. Enumerating the at most $|\{d_{h_1,h_2}\}|$ distinct thresholds yields an exact polynomial-time separation routine for the complete family. The anchor-based procedure of Algorithm 1 is a static restriction of this scheme: it generates at most $|H(l_1)| \cdot |H(l_2)|$ deduplicated patterns per pair in $O(|H(l_1)|^2 |H(l_2)|^2)$ time without consulting a fractional point, and posts them once before the solve. We adopt the static variant because the deduplicated pattern count is small on our benchmarks, whose distance matrices contain at most 13 distinct values, and because the resulting cuts already give robust exact performance in the computational study. Whether the biclique inequalities define facets of the underlying polytope, how they relate formally to RLT and lift-and-project constraints, and how they compare with semidefinite QAP relaxations remain open questions that we leave for future research.

3.4. SameRoom and SameAttendees side constraints

Capacity limits, mandatory hall features, and soft per-assignment preferences are already part of the base model through the compatibility set $H(l)$ and the penalty coefficients p_{lh} defined in Section 2. The timetabling literature also considers pairwise side constraints that couple the hall choices of two different lectures and cannot be encoded through $H(l)$ or p_{lh} alone. This subsection incorporates the two most common such constraints from the ITC 2019 competition model (Müller et al., 2025), *SameRoom* and *SameAttendees*, in both hard and soft forms.

The *SameRoom* constraint requires two lectures, scheduled at non-overlapping times, to be assigned to the same hall. Let C_1^{hard} be the set of lecture pairs for which a hard SameRoom constraint is imposed. Because equality is transitive, for each connected component of the

graph (L, C_1^{hard}) first replace the compatibility set of every lecture in the component by the intersection of all compatibility sets in that component; an empty intersection certifies infeasibility. On these reduced sets, impose

$$x_{l_1,h} = x_{l_2,h} \qquad \forall (l_1, l_2) \in C_1^{hard}, h \in H(l_1) \cap H(l_2). \tag{13}$$

The component-wide compatibility intersection makes (13) sufficient to enforce a common hall. Equivalently, one may retain the original sets, impose (13) on the pairwise intersection, and fix to zero the assignment variables indexed by either side of the symmetric difference; this is the encoding used in the implementation.

Alternatively, suppose that SameRoom is a soft constraint for a pair of lectures l_1, l_2, with penalty π_{l_1,l_2} if the two lectures are assigned to different halls. Let C_1^{soft} be the set of such pairs, and for each pair introduce a binary variable v_{l_1,l_2} that equals 1 when the constraint is violated. A linear encoding for the MIP formulation is

$$v_{l_1,l_2} \geq x_{l_1,h} - x_{l_2,h} \qquad \forall (l_1, l_2) \in C_1^{soft},\ h \in H(l_1) \cap H(l_2), \tag{14}$$

$$v_{l_1,l_2} \geq x_{l_2,h} - x_{l_1,h} \qquad \forall (l_1, l_2) \in C_1^{soft},\ h \in H(l_1) \cap H(l_2), \tag{15}$$

$$v_{l_1,l_2} \geq x_{l_1,h} \qquad \forall (l_1, l_2) \in C_1^{soft},\ h \in H(l_1) \setminus H(l_2), \tag{16}$$

$$v_{l_1,l_2} \geq x_{l_2,h} \qquad \forall (l_1, l_2) \in C_1^{soft},\ h \in H(l_2) \setminus H(l_1). \tag{17}$$

The corresponding penalty term $\sum_{(l_1,l_2)\in C_1^{soft}} \pi_{l_1,l_2} v_{l_1,l_2}$ is then added to the objective function. In the MIQP formulation, one may avoid the auxiliary variables and add the direct bilinear penalty

$$\sum_{(l_1,l_2)\in C_1^{soft}} \pi_{l_1,l_2} \left(1 - \sum_{h\in H(l_1)\cap H(l_2)} x_{l_1 h} x_{l_2 h}\right). \tag{18}$$

Because each lecture is assigned to exactly one hall, the inner sum equals 1 if and only if the two lectures use the same compatible hall, and equals 0 otherwise, so this term is exactly the SameRoom violation penalty.

SameRoom constraints, in both their hard and soft forms, may link lectures on different days. For example, a Monday lecture may be required to use the same hall as a related Wednesday lecture. Such a constraint breaks the daily decomposition because Monday's hall choice constrains Wednesday's; the coupled week, or possibly the full term, must then be solved jointly.

The *SameAttendees* constraint is also relevant to fixed-time hall assignment (Müller et al., 2025). In the timetabling model, it limits the travel time between two lectures that may share students. Once event times are fixed, it restricts the admissible hall pairs only when the time separation makes walking time relevant. A SameAttendees pair on different days imposes no hall-pair restriction under the usual travel-time interpretation and therefore creates no cross-day coupling. Let C_2^{hard} be the set of lecture pairs with a hard SameAttendees constraint, and let $F_{l_1,l_2} \subseteq H(l_1) \times H(l_2)$ denote the set of forbidden hall pairs for a given pair (l_1, l_2). If l_1 ends before l_2 starts, then $(h_1, h_2) \in F_{l_1,l_2}$ whenever

$$d_{h_1,h_2} > \text{start}(l_2) - \text{end}(l_1),$$

with the symmetric definition when l_2 ends before l_1 starts; if the two lectures overlap in time, then every hall pair is forbidden. The hard constraint is therefore expressed by

$$x_{l_1,h_1} + x_{l_2,h_2} \leq 1 \qquad \forall (l_1, l_2) \in C_2^{hard},\ (h_1, h_2) \in F_{l_1,l_2}. \tag{19}$$

Now let C_2^{soft} be the set of soft SameAttendees pairs, with penalty ρ_{l_1,l_2} for violating pair (l_1, l_2). Introduce a binary variable u_{l_1,l_2} that equals 1 when the chosen hall pair belongs to F_{l_1,l_2}. Then the MIP model can enforce

$$u_{l_1,l_2} \geq x_{l_1,h_1} + x_{l_2,h_2} - 1 \qquad \forall (l_1, l_2) \in C_2^{soft},\ (h_1, h_2) \in F_{l_1,l_2}, \tag{20}$$

Table 3
Equations that encode each hard/soft side constraint in each formulation.

Side constraint	MIQP	MIP
SameRoom, hard	(13)	(13)
SameRoom, soft	(18)	(14)–(17)
SameAttendees, hard	(19)	(19); strengthened by (22)
SameAttendees, soft	(21)	(20); strengthened by (23)

and add the penalty term $\sum_{(l_1,l_2)\in C_2^{soft}} \rho_{l_1,l_2} u_{l_1,l_2}$ to the objective. In the MIQP formulation, the same penalty can be written directly as

$$\sum_{(l_1,l_2)\in C_2^{soft}} \rho_{l_1,l_2} \sum_{(h_1,h_2)\in F_{l_1,l_2}} x_{l_1,h_1} x_{l_2,h_2}. \tag{21}$$

Because exactly one hall pair is selected for (l_1, l_2), the inner sum is a binary indicator of whether the chosen pair belongs to F_{l_1,l_2}.

In the MIP model, the pairwise formulations above can be strengthened using another set of biclique constraints. Let $S_1 \subseteq H(l_1)$ and $S_2 \subseteq H(l_2)$ be such that $S_1 \times S_2 \subseteq F_{l_1,l_2}$. Then every assignment that places l_1 in a hall from S_1 and l_2 in a hall from S_2 necessarily violates SameAttendees. Hence the following aggregated inequalities are valid (see Eqs. (22) and (23) in Box I): These inequalities dominate the basic pairwise constraints whenever $|S_1| > 1$ or $|S_2| > 1$. In the implementation, the extended cut mode constructs such sets by starting from one anchor hall of l_1, collecting all halls of l_2 that are forbidden with that anchor, and then enlarging the l_1-side set to every hall that is incompatible with the entire resulting set on the l_2 side. This is directly analogous in spirit to the extended distance cuts used for the quadratic walking term. These aggregated inequalities are useful for the linearized MIP; in the MIQP formulation, the soft SameAttendees penalties are represented directly by bilinear objective terms instead.

Table 3 summarizes, for each of the four hard/soft variants of the two side constraints, the equation number that encodes it in each formulation.

4. Numerical experiments

This section reports the numerical evaluation of the proposed formulations and modeling improvements on the real-world benchmark. Section 4.1 describes how the daily instances are extracted from the ITC 2019 and Lancaster 2023 data sources. Section 4.2 then summarizes the 1800-second exact computational campaign, compares the MIQP and MIP formulations, studies the contribution of biclique strengthening and capacity-dominance cuts, and quantifies how much the optimized assignment reduces model-implied student walking relative to the deployed Lancaster room plan.

The MIQP and MIP formulations were implemented in Python using Gurobi 13.0. The same Python code base handles data preparation and cut generation, and the experimental campaigns were executed by shell scripts. For reproducibility, the code, its documentation, and detailed result files are available at https://github.com/tal69/lecture-halls; the exact submitted-code release, version v1.0.3, is archived at https://doi.org/10.5281/zenodo.21310605. All experiments were run on an AMD Ryzen 9 5950X with 128 GB of RAM under Ubuntu 22.04, using 12 solver threads per run.

4.1. Test instances

Our real-world benchmark is derived from the ITC 2019 university course timetabling repository (Müller et al., 2025) and from a 2023 institutional dataset from Lancaster University, kindly provided by Matthew Davison, the first author of Davison et al. (2025). In both sources we retain the original hall set, hall capacities, and hall-to-hall distances, and transform a weekly timetable into independent daily lecture-hall assignment instances. This extraction retains all side

$$\sum_{h_1 \in S_1} x_{l_1,h_1} + \sum_{h_2 \in S_2} x_{l_2,h_2} \le 1 \qquad \forall (l_1, l_2) \in C_2^{hard},\ S_1 \times S_2 \subseteq F_{l_1,l_2}, \tag{22}$$

$$u_{l_1,l_2} \ge \sum_{h_1 \in S_1} x_{l_1,h_1} + \sum_{h_2 \in S_2} x_{l_2,h_2} - 1 \qquad \forall (l_1, l_2) \in C_2^{soft},\ S_1 \times S_2 \subseteq F_{l_1,l_2}. \tag{23}$$

Box I.

constraints whose lectures both belong to the selected day. Thus the reported experiments enforce SameRoom and SameAttendees constraints that survive the daily extraction, but they do not test cross-day SameRoom constraints, which would require the weekly or term-level model discussed in Section 3.4. For each source we first select the most loaded teaching week, where load is measured by the number of active hall-requiring lectures after input preparation. We then retain its five working days and discard weekend activity.

The ITC 2019 repository contains 36 real-world timetabling instances, namely 30 competition instances and 6 test instances (Müller et al., 2025). For the main large-scale experiments we use the five competition instances with the largest numbers of source classes among those that simultaneously contain a nontrivial hall-distance matrix and explicit student-enrollment data. For each selected ITC source we use a public timetable posted in the ITC 2019 repository as the fixed weekly schedule from which daily hall-assignment instances are extracted. After selecting the peak week, we keep its five working days and drop the weekend, so each ITC source contributes five daily instances. According to the official competition results posted on the ITC 2019 website (Müller et al., 2020), the winning submissions for the selected instances were by Dennis S. Holm, Rasmus Ø. Mikkelsen, Matias Sørensen, and Thomas R. Stidsen for `muni-pdf-spr16c`, `pu-proj-fal19`, `muni-pdfx-fal17`, and `pu-d9-fal19`, and by Karim Er-rhaimini for `agh-fal17`. Both winning approaches are documented in solver papers published on the contest website (Holm et al., 2020; Er-rhaimini, 2020). Table 4 summarizes the selected source instances and the resulting benchmark sizes.

For Lancaster University, we use an anonymized XML approximation derived from the actual 2023/24 timetable and student-registration data (Davison, 2025a,b). We are grateful to Matthew Davison for making these data available. The Lancaster XML contains *SameClass* groups, that is, sets of lecture identifiers that are administratively distinct but correspond to the same teaching activity and must therefore share the same time and hall. Our input-preparation pipeline contracts each *SameClass* component before any hall-assignment instance is created. It then identifies the two main teaching terms and selects the most loaded week within each term; in the present data, these are weeks 13 and 28. For each selected term week we again retain the five working days and discard weekend activity, so the Lancaster source contributes ten daily instances overall. A small number of registrations that violate the single-lecture-at-a-time assumption of Section 2 are removed from one of the two overlapping lectures, so that the cleaned data satisfy the assumption and the induced student-movement relation is well defined for the quadratic objective. After this cleaning, the hard and soft side constraints are inherited from the XML through the merged components; in the resulting Lancaster instances there are hard and soft *SameAttendees* constraints but no remaining *SameRoom* constraints.

Three aspects of the Lancaster data pipeline deserve explicit characterization, since they affect the construction of the inputs d_{h_1,h_2} and c_{l_1,l_2} used in the optimization models. First, the campus geometry is not supplied as a set of two-dimensional coordinates. Instead, the Lancaster dataset encodes the campus as a sparse, symmetric travel-time matrix embedded directly in the ITC 2019 XML format (Davison, 2025a,b; Müller et al., 2025). Each entry d_{h_1,h_2} is a nonnegative integer representing the number of five-minute time slots required to walk from hall h_1 to hall h_2; the granularity arises from the XML slot resolution of 288 slots per day, i.e., $1440/288 = 5$ min per slot. The 660 halls produce six distinct positive travel values $d_{h_1,h_2} \in \{1, 2, 3, 4, 5, 7\}$ (corresponding to 5, 10, 15, 20, 25, and 35 min of walking time), with the most common values being 1 and 2. Pairs of halls for which no `<travel>` entry appears in the XML are treated as co-located ($d_{h_1,h_2} = 0$); no hall features are defined, so the compatibility sets $H(l)$ are driven entirely by hall capacities.

Second, instead of using a published timetable solution as a fixed schedule (the approach used for the ITC instances), the Lancaster pipeline reads the zero-penalty time assignment embedded in the XML class definitions, which records the schedule that was actually used by the institution. The zero-penalty room option likewise records the allocated location in that schedule. Since each class has exactly one zero-penalty time option, the timetable is unique and deterministic: the SameClass-contracted merged components inherit their day, start, and duration fields directly from this institutional assignment. Third, the shared-enrollment weight c_{l_1,l_2} is constructed from individual student registrations, which are listed explicitly in the XML. Because many Lancaster courses offer multiple teaching configurations, the pipeline performs a greedy student-level assignment: for each student, courses are sorted by the number of feasible weekly-meeting options and each course is assigned to the most active feasible configuration that does not conflict with the student's already-scheduled components and does not exceed the hall capacity implied by the merged-component's assigned room. After this greedy pass, a student contributes to c_{l_1,l_2} if they attend both lectures l_1 and l_2 and the two lectures are immediate successors on the same day, that is, lecture l_2 starts within a short-break gap (automatically inferred from the most common positive inter-lecture gap in the data) after lecture l_1 ends.

Overall, the benchmark contains 35 daily instances: 25 from the five selected ITC source XML files and 10 from Lancaster. The ITC subset combines five large competition benchmarks, whereas the Lancaster subset contributes a much larger hall inventory and a different construction pipeline based on cleaned institutional registrations. Together, these sources provide a useful test bed for assessing the robustness of the MIQP and MIP formulations across materially different levels of hall abundance, timetable density, and student-flow structure. Side constraints are enforced whenever both lectures of a pair belong to the same daily instance: the ITC instances carry within-day hard and soft *SameRoom* and *SameAttendees* pairs, while the Lancaster instances carry hard and soft *SameAttendees* pairs only; their *SameRoom* requirements have been absorbed by the *SameClass* contraction. Pairs whose lectures fall on different days are necessarily dropped by the daily extraction; in particular, cross-day *SameRoom* requirements, which would couple the daily subproblems (Section 3.4), are not represented in this benchmark, and assessing the scalability of the formulations on weekly instances with such coupling is left for future work.

Both data sources have previously appeared in studies that optimize broader timetabling or hall-allocation objectives without the quadratic walking term considered here, so they do not provide like-for-like algorithmic benchmarks. The deployed Lancaster room plan instead

Table 4
Source instances underlying the real-world benchmark after retaining the five working days of the selected weeks, sorted in ascending order by number of halls. The column $|d|$ is the number of distinct values in the hall-to-hall distance matrix (including zero for self-distance and co-located pairs). The five ITC rows yield 25 daily lecture-hall assignment instances in total; the Lancaster row summarizes the two main 2023 teaching terms and yields 10 additional daily instances. A student is counted once if they appear in at least one retained day of the corresponding source, whereas registrations are summed over all retained days.

Source XML	Description	Halls	$\|d\|$	Sel. wk(s)	Lect./day	Students	Regs.
`muni-pdf-spr16c`	Masaryk, Spring 2016	70	9	3	159–222	1,715	9,869
`muni-pdfx-fal17`	Masaryk, Fall 2017	86	9	12	197–388	5,083	38,256
`pu-d9-fal19`	Purdue, Fall 2019	224	10	2	846–977	35,014	286,385
`agh-fal17`	AGH UST, Fall 2017	327	12	4	537–558	6,192	78,854
`lancs-yr23`	Lancaster, 2023 T1–T2	660	7	13, 28	306–597	13,274	196,173
`pu-proj-fal19`	Purdue, Fall 2019	770	13	6	2208–2562	38,248	437,517

Table 5
Best performance of each formulation family over the four cut configurations under an 1800-second solver limit. "Best incumbent" counts instances matching the smallest objective found by either formulation. "Median proof time" is the median, across solved instances, of the family's fastest end-to-end optimal run.

Source	Formulation	Instances	Optimal	Best incumbent	Median proof time (s)
ITC 2019	MIQP	25	20	23	0.78
ITC 2019	MIP	25	25	25	2.20
Lancaster	MIQP	10	10	10	51.57
Lancaster	MIP	10	10	10	18.11

provides an operational baseline for estimating the potential reduction relative to current practice; this comparison is reported in Section 4.2.4.

4.2. Analysis of numerical results

4.2.1. Experimental design and formulation comparison

To evaluate the two formulations, we solved each of the 35 instances from Section 4.1 with MIQP and MIP under all four combinations of the two modeling improvements from Section 3.3. This yields 280 exact runs under an 1800-second solver limit. To isolate the effect of the strengthening devices on the compact formulation's continuous relaxation, we also ran 140 diagnostic root-node solves of the linearized MIP under the same instance and enhancement combinations.

Table 5 summarizes the best result each formulation family attains over its four configurations. The compact MIP is the more robust exact method: at least one MIP configuration proves optimality on every one of the 35 daily instances, including all five large `pu-proj` days. The direct MIQP proves optimality on all 10 Lancaster days and on 20 of the 25 ITC days. Across its four configurations, MIQP matches the best incumbent on 33 of the 35 instances. It misses `pu-proj w6d1` by 8 objective units (0.033% of the optimum) and `pu-proj w6d3` by 13 units (0.053%). Method comparisons use end-to-end wall time, including cut generation, model construction, and the solver call. The 1800-second limit applies to the solver call, so end-to-end time can exceed it.

4.2.2. Fastest method combinations

The role of the modeling improvements is most visible when examining the fastest individual configurations. Table 6 reports, for each daily instance, the three fastest configurations that attain the reported UB and terminate with solver status `OPTIMAL` under the solver's numerical tolerance. Method codes have the form *formulation–B–D*: B denotes biclique distance strengthening and $\bar{B}$ its absence, while D denotes capacity-dominance cuts and $\bar{D}$ their absence. On the smaller and easier ITC days, the direct MIQP without biclique cuts is often fastest; on the hard Purdue instances and most Lancaster days, biclique-strengthened MIP configurations dominate the proof times and are the only configurations that solve all instances reliably.

Aggregating the same runs by method combination gives the complementary view in Table 7. The only individual configuration that proves optimality on all 35 instances is the biclique-strengthened compact formulation MIP–B–D.

Tables 6 and 7 also show the transition between easy and hard regimes. On the AGH, `muni-pdf`, and `muni-pdfx` days, the direct MIQP without biclique cuts often gives the lowest end-to-end wall time because these instances are already easy and the overhead of cut generation and model construction can dominate. On the harder Purdue days and most Lancaster days, biclique-strengthened MIP configurations provide the fastest proofs and are the only configurations that solve all instances reliably. The hardest Purdue project days remain nontrivial even after strengthening: `pu-proj w6d1` and `pu-proj w6d3` require about 27 and 25 min, respectively, for the fastest end-to-end proof. Those are precisely the cases on which the direct MIQP fails to prove optimality, and in several MIQP configurations with biclique cuts it returns no incumbent at all within the time limit. Thus the direct MIQP is effective on easy instances and for incumbent generation, whereas the compact MIP provides reliable optimality proofs across the benchmark.

4.2.3. Effects of the enhancements

Biclique strengthening also dominates at the root-relaxation level. In the diagnostic root-node experiment, the root lower bound of the compact MIP was compared with the final certified optimum for the corresponding instance, using the relative gap $(z^\star - \mathrm{LB}_{\mathrm{root}})/z^\star$. Across the 70 no-biclique root runs, the mean root gap was 9.91% and the median was 8.53%; 6 no-biclique runs, all on ITC instances, reached the 1800-second limit before completing the root. With biclique cuts enabled, no root run reached the time limit, the mean root gap fell to 0.14%, and the median root gap was zero. In paired comparisons holding the instance and capacity-dominance setting fixed, adding biclique cuts improved the bound in 47 of 70 pairs and reduced the root gap by 9.77 percentage points on average. These root results attribute part of the exact-performance gains in Tables 6 and 7 to the substantially tighter initial relaxation.

The capacity-dominance inequalities have a smaller aggregate effect. On the hardest instances they are part of the fastest configuration, for example on `pu-proj w6d1`, `pu-proj w6d3`, and `pu-proj w6d5`, and they are needed by the only configuration that proves optimality on all 35 instances. They are absent from the fastest configuration on many other days. This is consistent with the structure of the compatibility sets: many incompatible lecture–hall pairs are excluded for non-capacity reasons, so a cut family based on capacity thresholds has less leverage than the distance bicliques.

4.2.4. Value of optimization

We quantify the potential improvement relative to current practice on the ten Lancaster days, whose *status-quo* assignment uses the rooms actually allocated by the institution and is therefore an operational baseline; student travel between rooms is a recognized scheduling consideration in this setting (Davison et al., 2025). The student-flow weights are reconstructed from registration data by the greedy procedure described in Section 4.1, so the resulting walking totals are model-implied burdens rather than observed pedestrian trajectories.

Table 6
Three fastest exact methods for each daily instance under the 1800-second solver limit. Every listed run attains the reported UB and has status OPTIMAL under Gurobi's default 0.01% relative MIP-gap tolerance. Each ranking cell gives the method followed by its end-to-end time in seconds in parentheses. A dash in the second or third column indicates that no configuration at that rank both attained the reported UB and proved optimality.

Instance	$\|L\|$	UB	Fastest (s)	Second (s)	Third (s)
agh w4d1	542	966	MIQP–$\bar{B}$–$\bar{D}$ (0.73)	MIQP–$\bar{B}$–D (0.84)	MIP–$\bar{B}$–D (0.84)
agh w4d2	558	649	MIQP–$\bar{B}$–$\bar{D}$ (0.55)	MIP–$\bar{B}$–$\bar{D}$ (0.63)	MIQP–$\bar{B}$–D (0.69)
agh w4d3	540	509	MIQP–$\bar{B}$–$\bar{D}$ (0.42)	MIP–$\bar{B}$–$\bar{D}$ (0.45)	MIQP–$\bar{B}$–D (0.46)
agh w4d4	537	523	MIQP–$\bar{B}$–$\bar{D}$ (0.56)	MIQP–$\bar{B}$–D (0.63)	MIP–B–$\bar{D}$ (0.95)
agh w4d5	540	559	MIQP–$\bar{B}$–$\bar{D}$ (0.42)	MIQP–$\bar{B}$–D (0.49)	MIP–$\bar{B}$–$\bar{D}$ (0.55)
muni-pdf w3d1	217	85	MIP–B–$\bar{D}$ (0.00)	MIP–$\bar{B}$–$\bar{D}$ (0.01)	MIP–$\bar{B}$–D (0.01)
muni-pdf w3d2	222	143	MIQP–$\bar{B}$–$\bar{D}$ (0.00)	MIP–$\bar{B}$–$\bar{D}$ (0.00)	MIP–B–$\bar{D}$ (0.00)
muni-pdf w3d3	217	148	MIQP–$\bar{B}$–$\bar{D}$ (0.00)	MIP–$\bar{B}$–$\bar{D}$ (0.00)	MIP–B–$\bar{D}$ (0.00)
muni-pdf w3d4	192	120	MIQP–$\bar{B}$–$\bar{D}$ (0.00)	MIP–$\bar{B}$–$\bar{D}$ (0.00)	MIP–B–$\bar{D}$ (0.00)
muni-pdf w3d5	159	306	MIQP–$\bar{B}$–$\bar{D}$ (0.66)	MIQP–$\bar{B}$–D (0.72)	MIP–$\bar{B}$–$\bar{D}$ (1.08)
muni-pdfx w12d1	361	1611	MIQP–$\bar{B}$–D (1.27)	MIQP–$\bar{B}$–$\bar{D}$ (1.32)	MIP–B–$\bar{D}$ (2.02)
muni-pdfx w12d2	356	1543	MIQP–$\bar{B}$–$\bar{D}$ (1.14)	MIQP–$\bar{B}$–D (1.20)	MIP–$\bar{B}$–D (3.60)
muni-pdfx w12d3	388	2365	MIQP–$\bar{B}$–$\bar{D}$ (1.41)	MIQP–$\bar{B}$–D (1.66)	MIP–B–$\bar{D}$ (2.20)
muni-pdfx w12d4	332	1469	MIQP–$\bar{B}$–$\bar{D}$ (1.10)	MIQP–$\bar{B}$–D (1.12)	MIP–$\bar{B}$–$\bar{D}$ (2.44)
muni-pdfx w12d5	197	909	MIQP–$\bar{B}$–$\bar{D}$ (0.82)	MIQP–$\bar{B}$–D (0.94)	MIP–$\bar{B}$–$\bar{D}$ (1.18)
pu-d9 w2d1	909	12302	MIP–B–D (49.67)	MIP–B–$\bar{D}$ (50.45)	–
pu-d9 w2d2	846	6066	MIP–B–$\bar{D}$ (2.63)	MIP–B–D (2.81)	MIQP–$\bar{B}$–$\bar{D}$ (3.12)
pu-d9 w2d3	977	12655	MIP–B–$\bar{D}$ (186.26)	MIP–B–D (363.43)	–
pu-d9 w2d4	857	4902	MIP–B–D (11.53)	MIQP–$\bar{B}$–$\bar{D}$ (16.57)	MIP–B–$\bar{D}$ (18.71)
pu-d9 w2d5	883	10945	MIP–B–$\bar{D}$ (36.62)	MIP–B–D (40.19)	–
pu-proj w6d1	2348	24314	MIP–B–D (1632.58)	–	–
pu-proj w6d2	2362	10078	MIP–B–$\bar{D}$ (34.87)	MIP–B–D (38.25)	MIQP–B–D (710.44)
pu-proj w6d3	2562	24651	MIP–B–D (1513.00)	MIP–B–$\bar{D}$ (1562.31)	–
pu-proj w6d4	2320	9980	MIP–B–$\bar{D}$ (22.92)	MIP–B–D (24.36)	MIQP–B–D (363.28)
pu-proj w6d5	2208	21373	MIP–B–D (223.15)	MIP–B–$\bar{D}$ (268.21)	–
lancs t1 w13d1	528	4319	MIP–B–$\bar{D}$ (11.91)	MIP–B–D (12.26)	MIQP–B–D (39.38)
lancs t1 w13d2	528	2585	MIP–B–$\bar{D}$ (8.39)	MIP–B–D (8.61)	MIQP–$\bar{B}$–$\bar{D}$ (22.60)
lancs t1 w13d3	306	1650	MIQP–$\bar{B}$–D (6.12)	MIQP–$\bar{B}$–$\bar{D}$ (6.93)	MIP–B–$\bar{D}$ (84.69)
lancs t1 w13d4	587	2848	MIP–B–$\bar{D}$ (18.67)	MIP–B–D (18.77)	MIQP–B–$\bar{D}$ (56.85)
lancs t1 w13d5	510	2866	MIP–B–D (89.58)	MIP–B–$\bar{D}$ (93.22)	MIQP–B–D (132.65)
lancs t2 w28d1	548	3105	MIQP–$\bar{B}$–$\bar{D}$ (46.29)	MIQP–$\bar{B}$–D (46.76)	MIP–B–$\bar{D}$ (68.87)
lancs t2 w28d2	533	3944	MIP–B–$\bar{D}$ (28.83)	MIP–B–D (33.51)	MIQP–B–$\bar{D}$ (87.53)
lancs t2 w28d3	312	1860	MIP–B–$\bar{D}$ (7.70)	MIP–B–D (7.78)	MIQP–$\bar{B}$–D (11.92)
lancs t2 w28d4	597	2673	MIP–B–$\bar{D}$ (17.54)	MIP–B–D (17.88)	MIQP–B–D (57.71)
lancs t2 w28d5	546	2772	MIP–B–D (9.64)	MIP–B–$\bar{D}$ (11.42)	MIQP–$\bar{B}$–D (62.84)

Table 7
Method-level counts over the 35 daily instances for the eight MIQP/MIP configurations. "Found optimum" counts instances in which the method either returned a solution whose objective equals the established optimal value or proved optimality under the solver's numerical tolerance. "Proved optimality" counts instances in which the solver terminated with status OPTIMAL. "Fastest optimum" counts instances on which the method has the smallest end-to-end wall time among the runs that found the optimum. Mean time is the corresponding end-to-end mean over all 35 runs.

Method	Found optimum	Proved optimality	Fastest optimum	Mean time (s)
MIQP–$\bar{B}$–$\bar{D}$	33	28	14	440.00
MIQP–$\bar{B}$–D	33	28	2	443.28
MIQP–B–$\bar{D}$	29	29	0	373.90
MIQP–B–D	30	30	0	367.69
MIP–$\bar{B}$–$\bar{D}$	28	19	0	915.26
MIP–$\bar{B}$–D	27	18	0	922.72
MIP–B–$\bar{D}$	**35**	34	12	125.61
MIP–B–D	**35**	**35**	7	**122.13**

The paper comparison excludes the ITC instances because their fixed schedules are winning competition solutions rather than deployed room plans; the replication script reports the analogous ITC figures. On every Lancaster day the status-quo assignment is feasible for the complete hard model: each recorded hall is compatible, no two lectures in the same hall overlap, and every hard *SameAttendees* constraint is satisfied. The comparison therefore evaluates two assignments over a common feasible region.

Table 8 reports, for each Lancaster day, the walking component of the status-quo assignment and of runs attaining the smallest full-objective value in the 1800-second campaign. The selected optimized assignments reduce model-implied walking on every day, by 20.5% to 45.3%, with a mean of 31.9% and an aggregate reduction of 32.7% over the ten days. Thus, on an institutional dataset derived from registration records and a deployed room plan, the full optimization model reduces its model-implied walking component by roughly one third. Runs attaining the same best full-objective value can realize slightly different walking components because assignment penalties and soft side-constraint penalties can offset the difference. We report the smallest walking component observed among those runs; selecting the largest observed component instead changes each reported reduction by at most 0.31 percentage points. Because the model minimizes the combined objective rather than walking alone, these reductions are lower bounds on the maximum walking-only reductions obtainable over the same feasible region.

4.2.5. Scope and limitations

The computational study has two limitations. First, apart from the status-quo comparison of Section 4.2.4, it compares only the proposed formulations and enhancements; it does not benchmark dedicated QAP codes or decomposition-based timetabling heuristics. These methods address different models: standard QAP codes do not incorporate interval-overlap feasibility, restricted compatibility sets, assignment penalties, or the side constraints of Section 3.4, while timetabling heuristics optimize different objectives over different decision spaces (cf. Section 4.1). Constructing compatible algorithmic benchmarks

Table 8
Value of optimization on the ten Lancaster 2023 daily instances: model-implied walking under the institution's deployed room assignment and under an assignment attaining the best full-objective value. Each total is measured in student travel slots and equals $\sum c_{l_1,l_2} d_{h_1,h_2}$; one Lancaster distance unit corresponds to five minutes. For each instance, the optimized column uses the smallest walking component observed among best-objective runs. The status quo is feasible for the complete hard model. The final row aggregates over the ten days.

Instance	$\|L\|$	Status-quo walking	Optimized walking	Reduction
`lancs t1 w13d1`	528	5,633	4,116	26.9%
`lancs t1 w13d2`	528	3,759	2,417	35.7%
`lancs t1 w13d3`	306	1,969	1,565	20.5%
`lancs t1 w13d4`	587	4,034	2,653	34.2%
`lancs t1 w13d5`	510	3,706	2,699	27.2%
`lancs t2 w28d1`	548	4,536	2,875	36.6%
`lancs t2 w28d2`	533	5,198	3,766	27.5%
`lancs t2 w28d3`	312	2,396	1,768	26.2%
`lancs t2 w28d4`	597	4,020	2,462	38.8%
`lancs t2 w28d5`	546	4,681	2,561	45.3%
All ten days		39,932	26,882	32.7%

remains future work. Second, the study is confined to daily instances; weekly instances with cross-day *SameRoom* coupling are outside the benchmark, and their scaling behavior remains untested.

5. Conclusions and future research

The numerical results point to a clear hierarchy among the proposed enhancements. The compact linearized MIP is the most robust exact formulation on the benchmark: under an 1800-second solver limit, at least one MIP configuration solves all 25 ITC daily instances and all 10 Lancaster daily instances to proven optimality. The direct MIQP matches the best objective on 33 of the 35 instances, but it proves optimality on only 30 and is substantially slower on Lancaster. Biclique strengthening is the dominant modeling device on the difficult cases. On easy days, MIQP without biclique cuts often has the fastest proof because the proof itself is immediate and cut-generation overhead dominates; on the hard Purdue days and most Lancaster days, biclique-strengthened MIP configurations provide the fastest proofs and account for all configurations that solve every instance. Capacity-dominance cuts have a smaller and less uniform effect, but they are needed by the only configuration that proves optimality on all 35 instances.

During the revision, we also tested a constraint-programming formulation based on OR-Tools CP-SAT. In the same 1800-second campaign, it never produced a strictly better incumbent or the fastest proof of optimality and proved fewer instances than the compact MIP, so it was omitted from the core formulations and numerical comparison.

Taken together, these findings suggest that realistic lecture-hall assignment problems from both a competition-grade benchmark and an institutional dataset can be solved to optimality within practical time limits by off-the-shelf MIP technology once the biclique structure of the quadratic walking term is exploited. From a practical standpoint, this places the proposed framework within reach of operations offices that are already comfortable with mathematical-programming software, and it complements the multiobjective evolutionary approaches that target the closely related but broader hall-allocation problem (Antunes-Batista et al., 2026). The gains are concrete: on the ten Lancaster institutional days, optimization reduces model-implied walking by 20%–45% (about a third on average) relative to the university's deployed room plan, which is itself feasible for the complete hard model (Section 4.2.4).

Three directions for future research stand out. The first is the theoretical study of the biclique inequalities: identifying conditions under which they define facets of the underlying polytope, establishing formal dominance relations with RLT and lift-and-project constraints, comparing them with semidefinite relaxations of the walking term (Zhao et al., 1998; Anstreicher, 2003), and embedding their exact minimum-cut separation in a dynamic branch-and-cut scheme. The second is the integration of hall assignment with time scheduling under the walking-distance objective. Such integration is structurally challenging: (i) the immediate-successor relation, which is input data in our setting, becomes decision-dependent, since which lecture pairs are consecutive, and how many students they share, is itself determined by the time assignment, so the walking term involves products of time and hall decisions and becomes cubic in the joint binary variables, or requires an additional linearization layer over pair-consecutiveness indicators; (ii) the interval overlap graph and its clique structure, on which the non-overlap constraints and capacity-dominance cuts rely, are no longer fixed; and (iii) the daily decomposition is lost, because time decisions couple the days through instructor availability and curriculum constraints. A pragmatic intermediate step is therefore to allow limited schedule adjustments, such as swapping two lectures of the same course between adjacent time slots, which couples hall assignment with light timetable repair while preserving much of the structure exploited here. The third direction is the structural study of the problem beyond worst-case hardness: Section 2 shows that successor-graph sparsity alone does not yield tractability (Proposition 2) while bounded active-or-awaiting lecture concurrency does, and developing this into a systematic theory of tractable subclasses and fixed-parameter algorithms for the combined successor-overlap structure would complement the hardness results.

CRediT authorship contribution statement

Tal Raviv: Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Anthropic's Claude Cowork and OpenAI Codex to assist with manuscript revision, language editing, LaTeX editing, proof verification, code documentation, and code verification. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The complete code base, its documentation, and the detailed result files for the 280 Gurobi exact runs and 140 root-node diagnostic runs underlying the reported numerical campaign are publicly available at https://github.com/tal69/lecture-halls; release v1.0.3 is archived at https://doi.org/10.5281/zenodo.21310605. The ITC 2019 source XML files are publicly distributed through the ITC 2019 repository of Müller et al. (2025). The Lancaster 2023 source dataset is openly available under a CC BY license from Lancaster University at https://doi.org/10.17635/lancaster/researchdata/279.

References

Abdipoor, S., Yaakob, R., Goh, S.L., Abdullah, S., 2023. Meta-heuristic approaches for the university course timetabling problem. Intell. Syst. Appl. 19, 200253. http://dx.doi.org/10.1016/j.iswa.2023.200253.

Abdullah, S., Turabieh, H., 2012. On the use of multi-neighbourhood structures within a tabu-based memetic approach to university timetabling problems. Inform. Sci. 191, 146–168. http://dx.doi.org/10.1016/j.ins.2011.12.018.

Adams, W.P., Johnson, T.A., 1994. Improved linear programming-based lower bounds for the quadratic assignment problem. In: Quadratic Assignment and Related Problems. In: DIMACS Series in Discrete Mathematics and Theoretical Computer Science, vol. 16, American Mathematical Society, pp. 43–75.

Anstreicher, K.M., 2003. Recent advances in the solution of quadratic assignment problems. Math. Program. 97, 27–42. http://dx.doi.org/10.1007/s10107-003-0437-z.

Anstreicher, K., Brixius, N., Goux, J.P., Linderoth, J., 2002. Solving large quadratic assignment problems on computational grids. Math. Program. 91, 563–588. http://dx.doi.org/10.1007/s101070100260.

Antunes-Batista, B., Atta, S., Basto-Fernandes, V., Yevseyeva, I., Emmerich, M.T.M., 2026. Multiobjective optimization approaches for room allocation in university course timetabling. Procedia Comput. Sci. 278, 663–670. http://dx.doi.org/10.1016/j.procs.2026.03.037.

Bonutti, A., De Cesco, F., Di Gaspero, L., Schaerf, A., 2012. Benchmarking curriculum-based course timetabling: Formulations, data formats, instances, validation, visualization, and results. Ann. Oper. Res. 194, 59–70. http://dx.doi.org/10.1007/s10479-010-0707-0.

Burer, S., Vandenbussche, D., 2006. Solving lift-and-project relaxations of binary integer programs. SIAM J. Optim. 16, 726–750. http://dx.doi.org/10.1137/040609574.

Burkard, R.E., Çela, E., Pardalos, P.M., Pitsoulis, L.S., 1998. The quadratic assignment problem. In: Handbook of Combinatorial Optimization. Kluwer Academic Publishers, pp. 241–338.

Burke, E.K., Petrovic, S., 2002. Recent research directions in automated timetabling. European J. Oper. Res. 140, 266–280. http://dx.doi.org/10.1016/S0377-2217(02)00069-3.

Cardoen, B., Demeulemeester, E., Beliën, J., 2010. Operating room planning and scheduling: A literature review. European J. Oper. Res. 201, 921–932. http://dx.doi.org/10.1016/j.ejor.2009.04.011.

Carter, M.W., Tovey, C.A., 1992. When is the classroom assignment problem hard? Oper. Res. 40, S28–S39. http://dx.doi.org/10.1287/opre.40.1.S28.

Davison, M., 2025a. International timetabling competition 2019 approximation of Lancaster university 2023/2024 term. http://dx.doi.org/10.17635/lancaster/researchdata/279, Lancaster University Research Directory, dataset.

Davison, M., 2025b. Modelling and Solving Multi-Objective University Course Timetabling Problems (Ph.D. thesis). Lancaster University, http://dx.doi.org/10.17635/lancaster/thesis/3040.

Davison, M., Kheiri, A., Zografos, K.G., 2025. Modelling and solving the university course timetabling problem with hybrid teaching considerations. J. Sched. 28, 195–215. http://dx.doi.org/10.1007/s10951-024-00817-w.

Er-rhaimini, K., 2020. Forest growth optimization for solving timetabling problems. Solver summary paper on the official ITC 2019 website. URL: https://www.itc2019.org/papers/itc2019-er-rhaimini.pdf. (Accessed 17 April 2026).

Fischer, A., Fischer, F., Hungerländer, P., 2019. New exact approaches to row layout problems. Math. Program. Comput. 11, 703–754. http://dx.doi.org/10.1007/s12532-019-00162-6.

Frieze, A.M., Yadegar, J., 1983. On the quadratic assignment problem. Discrete Appl. Math. 5, 89–98. http://dx.doi.org/10.1016/0166-218X(83)90018-5.

Gilmore, P.C., 1962. Optimal and suboptimal algorithms for the quadratic assignment problem. J. Soc. Ind. Appl. Math. 10, 305–313. http://dx.doi.org/10.1137/0110022.

Golumbic, M.C., 2004. Algorithmic Graph Theory and Perfect Graphs, second ed. In: Annals of Discrete Mathematics, vol. 57, Elsevier, Amsterdam.

Haghani, A., Chen, M.C., 1998. Optimizing gate assignments at airport terminals. Transp. Res. A 32, 437–454. http://dx.doi.org/10.1016/S0965-8564(98)00005-6.

Hahn, P., Grant, T., 1998. Lower bounds for the quadratic assignment problem based upon a dual formulation. Oper. Res. 46, 912–922. http://dx.doi.org/10.1287/opre.46.6.912.

Holm, D.S., Mikkelsen, R.Ø., Sørensen, M., Stidsen, T.R., 2020. A mip based approach for international timetabling competition 2019. Solver summary paper on the official ITC 2019 website. URL: https://www.itc2019.org/papers/itc2019-holm.pdf. (Accessed 17 April 2026).

Kaufman, L., Broeckx, F., 1978. An algorithm for the quadratic assignment problem using Bender's decomposition. European J. Oper. Res. 2, 207–211. http://dx.doi.org/10.1016/0377-2217(78)90095-4.

Kolen, A.W.J., Lenstra, J.K., Papadimitriou, C.H., Spieksma, F.C.R., 2007. Interval scheduling: A survey. Naval Res. Logist. 54, 530–543. http://dx.doi.org/10.1002/nav.20231.

Koopmans, T.C., Beckmann, M., 1957. Assignment problems and the location of economic activities. Econometrica 25, 53–76. http://dx.doi.org/10.2307/1907742.

Lawler, E.L., 1963. The quadratic assignment problem. Manag. Sci. 9, 586–599. http://dx.doi.org/10.1287/mnsc.9.4.586.

Lewis, R., Paechter, B., 2005. Application of the grouping genetic algorithm to university course timetabling. In: Evolutionary Computation in Combinatorial Optimization. EvoCOP 2005, Springer, pp. 144–153. http://dx.doi.org/10.1007/978-3-540-31996-2_14.

Loiola, E.M., de Abreu, N.M.M., Boaventura-Netto, P.O., Hahn, P., Querido, T., 2007. A survey for the quadratic assignment problem. European J. Oper. Res. 176, 657–690. http://dx.doi.org/10.1016/j.ejor.2005.09.032.

McCollum, B., Schaerf, A., Paechter, B., McMullan, P., Lewis, R., Parkes, A.J., Di Gaspero, L., Qu, R., Burke, E.K., 2010. Setting the research agenda in automated timetabling: The second international timetabling competition. INFORMS J. Comput. 22, 120–130. http://dx.doi.org/10.1287/ijoc.1090.0320.

Müller, T., Rudová, H., Müllerová, Z., 2020. International timetabling competition 2019 awards slides. Official ITC 2019 website. URL: https://www.itc2019.org/papers/itc2019-awards-slides.pdf. (Accessed 17 April 2026).

Müller, T., Rudová, H., Müllerová, Z., 2025. Real-world university course timetabling at the international timetabling competition 2019. J. Sched. 28, 247–267. http://dx.doi.org/10.1007/s10951-023-00801-w.

Rappos, E., Thiémard, E., Robert, S., Hêche, J.F., 2022. A mixed-integer programming approach for solving university course timetabling problems. J. Sched. 25, 391–404. http://dx.doi.org/10.1007/s10951-021-00715-5.

Sahni, S., Gonzalez, T., 1976. P-complete approximation problems. J. ACM 23, 555–565. http://dx.doi.org/10.1145/321958.321975.

Schaerf, A., 1999. A survey of automated timetabling. Artif. Intell. Rev. 13, 87–127. http://dx.doi.org/10.1023/A:1006576209967.

Taillard, E., 1991. Robust taboo search for the quadratic assignment problem. Parallel Comput. 17, 443–455. http://dx.doi.org/10.1016/S0167-8191(05)80147-4.

Torres-Ovalle, C., Montoya-Torres, J.R., Quintero-Araújo, C.L., Sarmiento-Lepesqueur, A., Castilla-Luna, M., 2014. University course scheduling and classroom assignment. Ingeniería Y Universidad 18, 59–75. http://dx.doi.org/10.11144/Javeriana.IYU18-1.phaa.

Zhao, Q., Karisch, S.E., Rendl, F., Wolkowicz, H., 1998. Semidefinite programming relaxations for the quadratic assignment problem. J. Comb. Optim. 2, 71–109. http://dx.doi.org/10.1023/A:1009795911987.